Fiscal Year:	FY 2022	**Task Last Updated:**	FY 06/23/2022
PI Name:	Hirsa, Amir Ph.D.		
Project Title:	Amyloid Fibril Formation in Microgravity via Ring-Sheared Drop		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Rensselaer Polytechnic Institute		
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City:	Troy	**State:**	NY
Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	09/01/2013	**End Date:**	05/31/2023
No. of Post Docs:	3	**No. of PhD Degrees:**	4
No. of PhD Candidates:	10	**No. of Master' Degrees:**	7
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	12
No. of Bachelor's Candidates:	20	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Gorti, Sridhar	**Contact Phone:**	256 544-0158
Contact Email:	sridhar.gorti@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 05/31/2023 per NSSC information (Ed., 10/5/22) NOTE: End date changed to 9/30/2022 per continuation grant 80NSSC20K1726 issued, including title change now reflected in above Title, per S. Gorti/MSFC (Ed., 1/27/21) NOTE: Extended to 9/30/2020 per PI and NSSC information (Ed., 6/4/18)		
Key Personnel Changes/Previous PI:	June 2020 report: no changes		
COI Name (Institution):	Lopez, Juan Ph.D. (Arizona State University)		
Grant/Contract No.:	80NSSC20K1726 ; NNX13AQ22G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Ed. note January 2021: Continuation grant issued with slightly different title, Amyloid Fibril Formation in Microgravity via Ring-Sheared Drop 80NSSC20K1726 (formerly titled Amyloid Fibril Formation in Microgravity: Distinguishing Interfacial and Flow Effects NNX13AQ22G). Information combined with original grant NNX13AQ22G, per NASA Marshall Space Flight Center management] Formation of amyloid fibrils is widely studied because of their role in neurodegenerative diseases, including Alzheimer's. The fibrillization process is characterized by a nucleation step followed by fibril growth, much like the steps that occur during protein crystallization. Aside from the general interest in macromolecular biophysics, recent evidence shows the relevance of amyloid fibril formation in animal models exposed to cosmic radiation, with obvious implications for interplanetary travel. A preponderance of evidence shows that agitating protein solutions can significantly accelerate amyloid fibril formation; however, the mechanisms by which the presence of bubbles and flow independently affect fibrillization are not known. The situation is further complicated because we do not yet have mechanistic models that can quantify the influence of natural convection due to gravity on the fibrillization process. We know that fluid interfaces (e.g., air-water interface) induce amyloid fibrils when they are sheared, but the influence of container walls is not fully understood. Here, we aim to distinguish the influence of flow and fluid interfaces on amyloid formation by removing other effects, including solid walls and buoyancy effects. Formation of protein amyloid fibrils provides a model system for studying biomolecular transport phenomena, where the influence of flow can be altogether negated by the elimination of gravity. Furthermore, the experiments can be performed aboard the International Space Station (ISS) with essentially no solid container, since surface tension can provide liquid containment. Thus, the free surface can be readily observed (e.g., with high magnification objectives), manipulated (e.g., can be sheared or left otherwise undisturbed), and modified (e.g., by spreading a monomolecular oil film). First, we will perform experiments at 1g on a small droplet to ascertain the effect of forced convection on a liquid surface. Next, we will generalize our very successful computational model in annular systems to the spherical geometry of liquids at microgravity. We will design the space-based experiments based on the results from the ground experiments and numerical predictions. Finally, we will perform experiments aboard ISS on large-scale drops in an experiment referred to as the Ring-Sheared Drop (RSD).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	When proteins are perturbed from their native state, they lose their native confirmation (folding) and then can bond with other proteins, forming an aggregate. This is the reason why the transparent fluid in an egg becomes a white solid when cooked. Amyloid fibrils are a unique type of protein aggregate that occur naturally in the body. We study them because amyloid plaques in the brain are associated with many neurodegenerative disorders, including Alzheimer's. Fibrils are very intriguing because they are highly ordered and self-assemble. When heated, insulin molecules, for example, denature and bond with other denatured molecules (or assists in denaturing the other molecules). These form a nucleation site, and from there the fibril elongates as more molecule bond into what is called a "beta sheet" structure. The primary mechanism for activation of protein in the body is not yet understood, but it is likely to be a combination of flow friction, aka "shear stress," and interfacial effects that native proteins are exposed to. Aside from their central role in neurodegenerative diseases such as Parkinson's and Alzheimer's, formation of amyloid structures in proteins plays a role in other diseases including diabetes.
Task Progress:	Progress has been made in both scientific and engineering aspects of the project. 1) An experimental study on amyloidogenesis performed in the knife-edge (surface) viscometer, (KEV), an Earth analog of the Ring-Sheared Drop (RSD), where instead of surface tension, a cylindrical open dish is used for containment. 2) A combined experimental and computational study that examined the effects of microorganisms in centimeter scale drops in microgravity. This study was undertaken to assess a fundamental question about the utility of the RSD as a space-based bioreactor. 3) Amyloid fibril formation was successfully studied using the RSD aboard the International Space Station (ISS).
Bibliography Type:	Description: (Last Updated: 04/15/2026)
Articles in Peer-reviewed Journals	Adam JA, Middlestead HR, Debono NE, Hirsa AH. "Effects of shear rate and protein concentration on amyloidogenesis via interfacial shear." J. Phys. Chem. B. 2021 Sep 3;125(36):10355-10363. https://doi.org/10.1021/acs.jpcb.1c05171 , Sep-2021
Articles in Peer-reviewed Journals	McMackin PM, Adam JA, Griffin SR, Bonocora RP, Brakke KA, Lopez JM, Hirsa AH. "Effects of microorganisms on drop formation in microgravity during a parabolic flight with residual gravity and jitter." Microgravity Sci. Technol. 2022 Feb 22;34:15. https://doi.org/10.1007/s12217-022-09933-8 , Feb-2022
Articles in Peer-reviewed Journals	McMackin P, Adam J, Griffin S, Hirsa A. "Amyloidogenesis via interfacial shear in a containerless biochemical reactor aboard the International Space Station." npjMicrogravity. 2022 Sep 20;8(41). https://doi.org/10.1038/s41526-022-00227-2 ; PMID: 36127358; PMCID: PMC9489778 , Sep-2022